

[illegible]

1. Know the cause then pick the treatment

WHAT YOU SEE

Banner 'KNOW THE CAUSE THEN PICK THE TREATMENT'.

WHAT IT ENCODES

Polyuria/polydipsia management hinges on classifying the cause first: AVP deficiency (central), AVP resistance (nephrogenic), or primary polydipsia. Each has a distinct therapy — DDAVP, cause removal, vs water restriction.

BOARD TRAP

Treating before classifying is dangerous: DDAVP helps central DI but worsens primary polydipsia (hyponatremia) and is useless in nephrogenic DI.

2. AVP deficiency wing (central)

WHAT YOU SEE

Left wing 'AVP DEFICIENCY (formerly central)'.

WHAT IT ENCODES

Central DI causes cluster as trauma/surgery, tumor, infiltrative, infectious, and idiopathic/autoimmune. AVP synthesis or storage fails; serum Na trends high; treat with desmopressin.

BOARD TRAP

AVP deficiency = central DI, the DDAVP-responsive type — confusing the wing with resistance leads to the wrong drug.

3. Trauma / surgery

WHAT YOU SEE

Portrait 'TRAUMA / surgery' head-injured patient.

WHAT IT ENCODES

Head trauma or pituitary/hypothalamic surgery is a leading central DI cause; may follow a classic triphasic pattern (DI → SIADH → permanent DI). Monitor Na and urine output closely post-op.

BOARD TRAP

Post-neurosurgical patients can swing from DI to transient SIADH (hyponatremia) and back — fixed DDAVP dosing without Na monitoring causes dangerous swings.

4. Tumor

WHAT YOU SEE

Portrait 'TUMOR'.

WHAT IT ENCODES

Craniopharyngioma, germinoma, pituitary macroadenoma, and metastases (breast/lung) damage the hypothalamic-stalk axis to cause central DI. MRI localizes the lesion after copeptin confirms central type.

BOARD TRAP

A pituitary adenoma itself rarely causes DI; stalk/hypothalamic involvement (or post-op) does — pure anterior adenomas more often cause hormone excess/deficiency, not DI.

5. Infectious / infiltrative

WHAT YOU SEE

Bottom-left 'INFECTIOUS' portrait, infiltrative cues.

WHAT IT ENCODES

Infiltrative/granulomatous disease (sarcoidosis, Langerhans cell histiocytosis, tuberculosis, hypophysitis) and infections damage the hypothalamic-neurohypophyseal axis. Stalk thickening on MRI is a clue.

BOARD TRAP

Stalk thickening + DI in a young patient suggests histiocytosis/germinoma/sarcoid — don't dismiss isolated central DI as idiopathic without imaging.

6. AVP resistance wing (nephrogenic)

WHAT YOU SEE

Center 'AVP RESISTANCE (formerly nephrogenic)'.

WHAT IT ENCODES

Nephrogenic DI = kidney resistance to AVP from drugs, electrolytes, intrinsic renal disease, or congenital channel defects. Treat by removing the cause, low-solute diet, thiazide, and amiloride (for lithium).

BOARD TRAP

AVP resistance does NOT respond to DDAVP — the central drug fails here; the diagnostic clue is high baseline copeptin.

7. Lithium — most common acquired nephrogenic cause

WHAT YOU SEE

Central 'LITHIUM — MOST COMMON ACQUIRED CAUSE' pill bottle.

WHAT IT ENCODES

Chronic lithium accumulates in principal cells via ENaC, downregulating AQP2 to cause nephrogenic DI (often partial, sometimes irreversible). Amiloride blocks ENaC entry; thiazide reduces volume/urine output.

BOARD TRAP

In lithium NDI use amiloride to block ENaC uptake; never water-restrict a polyuric DI patient, and monitor lithium levels when adding a thiazide (toxicity risk).

8. Hypercalcemia / hypokalemia

WHAT YOU SEE

Center portraits with electrolyte cues.

WHAT IT ENCODES

Hypercalcemia and hypokalemia impair medullary concentrating gradient and AQP2, causing reversible acquired nephrogenic DI. Correcting the electrolyte restores concentrating ability.

BOARD TRAP

Electrolyte-induced nephrogenic DI is reversible — labeling it congenital/idiopathic misses an easy fix (normalize Ca/K).

9. Congenital — V2 receptor / AQP2 (sickle, etc.)

WHAT YOU SEE

Center sickle-cell and channel cues.

WHAT IT ENCODES

Congenital nephrogenic DI: X-linked V2-receptor mutation (most common, boys) or autosomal AQP2 defects. Sickle cell disease damages the medulla causing concentrating defects. Presents in infancy with failure to thrive.

BOARD TRAP

X-linked V2 mutation presents in male infants — a congenital cause is unresponsive to DDAVP, unlike acquired electrolyte causes that reverse.

10. Metabolic / intrinsic renal

WHAT YOU SEE

'METABOLIC' portrait, intrinsic-renal cues.

WHAT IT ENCODES

Intrinsic tubulointerstitial disease, obstruction relief, and amyloid impair the medullary gradient and AVP responsiveness, producing acquired nephrogenic DI. Often partial and accompanied by other tubular defects.

BOARD TRAP

Post-obstructive nephrogenic DI is self-limited tubular dysfunction — over-aggressive fluid replacement perpetuates the diuresis.

11. Primary polydipsia wing

WHAT YOU SEE

Right wing 'PRIMARY POLYDIPSIA'.

WHAT IT ENCODES

Primary polydipsia = excessive water intake (psychogenic, dipsogenic, or iatrogenic) suppressing AVP. Low-normal serum Na, dilute urine that concentrates with deprivation. Treat with fluid restriction, not DDAVP.

BOARD TRAP

Primary polydipsia presents with LOW serum Na (vs high-normal in DI) — giving DDAVP here precipitates severe hyponatremia.

12. Dipsogenic lesion

WHAT YOU SEE

Right wing 'dipsogenic lesion' brain portrait.

WHAT IT ENCODES

Dipsogenic polydipsia stems from a lowered thirst threshold (hypothalamic lesion, sarcoid, MS) driving excess drinking despite normal/low osmolality. Distinguish from psychogenic and from true DI.

BOARD TRAP

A reset thirst threshold mimics DI but the AVP/copeptin axis is intact — fluid restriction (carefully) helps, DDAVP harms.

13. Health-conscious / habitual high intake

WHAT YOU SEE

Right wing athletic figure with large water bottle.

WHAT IT ENCODES

Habitual high water intake (fitness 'drink more water' culture, medication regimens) can produce iatrogenic primary polydipsia with dilute polyuria and low-normal Na. History reveals the volume of intake.

BOARD TRAP

A well-appearing patient drinking liters 'to stay hydrated' with low-normal Na is primary polydipsia — don't escalate to DDAVP.

14. Compulsive water intake (psychogenic)

WHAT YOU SEE

Bottom 'COMPULSIVE WATER INTAKE' figure.

WHAT IT ENCODES

Psychogenic polydipsia is common in psychiatric patients (schizophrenia); compulsive water drinking can cause acute hyponatremia/water intoxication if intake outpaces excretion.

BOARD TRAP

Acute psychogenic water intoxication is a medical emergency (seizures) — the issue is excess intake, not AVP deficiency; restrict water, never give DDAVP.

15. Diagnostic lab first

WHAT YOU SEE

Bottom plaque 'POLYURIA POLYDIPSIA CAUSES HALL — see diagnostic lab first'.

WHAT IT ENCODES

Always classify with serum Na, urine osm, and copeptin testing BEFORE selecting therapy. The three wings demand opposite treatments, so misclassification is harmful.

BOARD TRAP

Skipping the lab/copeptin step and treating empirically risks giving the exactly wrong therapy for the patient's wing.
